MODEL LAB · 2026-10-27

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

17,710
held-out plate appearances

2.92%
held-out HR rate

0.6014
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,710 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5820	1.61×	0.02829	0.13108	0.00466
KNN comps Prior is the pooled rate of the nearest comparables.	0.6014	1.64×	0.02824	0.12993	0.00614
SVR Support-vector regression on the same feature matrix.	0.5940	1.51×	0.02829	0.13054	0.00689
Random forest Forest regression; a learned similarity metric.	0.5967	1.51×	0.02828	0.13022	0.00676
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6014	1.64×	0.02824	0.12993	0.00614
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6014	1.64×	0.02824	0.12993	0.00614
Linear regression Ordinary least squares on the same features.	0.5979	1.59×	0.02828	0.13031	0.00488
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5969	1.61×	0.02833	0.13060	0.00615
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5971	1.57×	0.02827	0.13021	0.00494
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5961	1.45×	0.02832	0.13060	0.00718

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6456	1.58×	0.02291	0.11126	0.01790
KNN comps Prior is the pooled rate of the nearest comparables.	0.6062	1.58×	0.02288	0.10980	0.01007
SVR Support-vector regression on the same feature matrix.	0.5783	1.58×	0.02323	0.11613	0.02309
Random forest Forest regression; a learned similarity metric.	0.6000	1.58×	0.02320	0.11319	0.02260
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6062	1.58×	0.02288	0.10980	0.01007
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6062	1.58×	0.02288	0.10980	0.01007
Linear regression Ordinary least squares on the same features.	0.6112	1.32×	0.02322	0.11383	0.02025
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6015	1.32×	0.02354	0.11442	0.02384
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.58×	0.02300	0.11128	0.01557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5934	1.05×	0.02361	0.11640	0.02762

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 17.2%, Junior Caminero 16.2%, William Contreras 16.0%, Jake Bauers 15.3%, Jonathan Aranda 13.3%, Brice Turang 13.3%
Flat league	5/6	Jackson Chourio 17.5%, William Contreras 15.8%, Junior Caminero 15.5%, Jake Bauers 14.2%, Brice Turang 13.5%, Yandy Díaz 13.5%
KNN comps	5/6	Jackson Chourio 17.8%, Junior Caminero 16.0%, William Contreras 15.9%, Jake Bauers 15.5%, Jonathan Aranda 13.4%, Yandy Díaz 13.2%
SVR	5/6	Jackson Chourio 18.2%, Junior Caminero 17.4%, William Contreras 15.9%, Jake Bauers 15.2%, Yandy Díaz 13.7%, Brice Turang 13.2%
Random forest	5/6	Jackson Chourio 18.1%, Junior Caminero 16.7%, William Contreras 15.8%, Jake Bauers 15.5%, Yandy Díaz 13.5%, Jonathan Aranda 13.0%
Core (KNN+SVR)	5/6	Jackson Chourio 17.8%, Junior Caminero 16.0%, William Contreras 15.9%, Jake Bauers 15.5%, Jonathan Aranda 13.4%, Yandy Díaz 13.2%
Forest blend	5/6	Jackson Chourio 17.8%, Junior Caminero 16.0%, William Contreras 15.9%, Jake Bauers 15.5%, Jonathan Aranda 13.4%, Yandy Díaz 13.2%
Linear regression	5/6	Jackson Chourio 18.0%, Junior Caminero 16.2%, William Contreras 15.9%, Jake Bauers 15.2%, Jonathan Aranda 13.3%, Yandy Díaz 13.3%
Logistic regression	5/6	Jackson Chourio 18.0%, Junior Caminero 16.6%, William Contreras 15.9%, Jake Bauers 15.3%, Yandy Díaz 13.2%, Jonathan Aranda 13.1%
Neural net (torch)	5/6	Jackson Chourio 18.1%, Junior Caminero 16.6%, William Contreras 16.0%, Jake Bauers 15.4%, Yandy Díaz 13.3%, Jonathan Aranda 13.2%
XGBoost	5/6	Jackson Chourio 18.3%, Junior Caminero 16.9%, William Contreras 15.8%, Jake Bauers 15.5%, Yandy Díaz 13.6%, Jonathan Aranda 13.1%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 Jackson Chourio MIL slate	**17.2%**	17.5%	17.8%	18.2%	18.1%	17.8%	17.8%	18.0%	18.0%	
2 Junior Caminero TB slate	**16.2%**	15.5%	16.0%	17.4%	16.7%	16.0%	16.0%	16.2%	16.6%	
3 William Contreras MIL	**16.0%**	15.8%	15.9%	15.9%	15.8%	15.9%	15.9%	15.9%	15.9%	
4 Jake Bauers MIL	**15.3%**	14.2%	15.5%	15.2%	15.5%	15.5%	15.5%	15.2%	15.3%	
5 Jonathan Aranda TB	**13.3%**	12.7%	13.4%	13.0%	13.0%	13.4%	13.4%	13.3%	13.1%	
6 Brice Turang MIL	**13.3%**	13.5%	13.0%	13.2%	12.8%	13.0%	13.0%	12.5%	12.4%	
7 Yandy Díaz TB	**13.1%**	13.5%	13.2%	13.7%	13.5%	13.2%	13.2%	13.3%	13.2%	
8 Christian Yelich MIL	**12.3%**	12.3%	12.6%	12.0%	12.3%	12.6%	12.6%	12.5%	12.0%	
9 Garrett Mitchell MIL	**12.2%**	11.7%	12.1%	11.3%	11.7%	12.1%	12.1%	11.6%	11.6%	
10 Victor Mesa Jr. TB	**10.9%**	10.6%	10.9%	11.2%	11.3%	10.9%	10.9%	11.0%	10.9%	
11 Jonny DeLuca TB	**8.0%**	8.7%	7.6%	7.5%	7.3%	7.6%	7.6%	7.7%	8.0%	
12 Liam Hicks TB	**7.7%**	8.6%	7.6%	8.4%	8.2%	7.6%	7.6%	8.0%	7.8%	
13 Cooper Pratt MIL	**7.2%**	9.2%	7.1%	6.7%	6.6%	7.1%	7.1%	6.1%	6.5%	
14 David Hamilton MIL	**6.8%**	8.1%	6.6%	6.1%	5.9%	6.6%	6.6%	6.4%	6.4%	
15 Sal Frelick MIL	**6.7%**	8.3%	6.9%	6.4%	6.1%	6.9%	6.9%	6.7%	6.7%	
16 Richie Palacios TB	**6.3%**	7.0%	6.2%	6.0%	5.5%	6.2%	6.2%	6.0%	6.0%	
17 Taylor Walls TB	**4.4%**	5.7%	4.2%	3.3%	3.3%	4.2%	4.2%	4.1%	4.2%	
18 Chandler Simpson TB	**2.6%**	5.7%	3.4%	1.6%	1.8%	3.4%	3.4%	1.5%	3.0%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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